한양대학교
HANYANG UNIVERSITY

Graduation Project Proposal

Title	Building an efficient distributed inference system for open-source LLMs on the cloud
Summary (in 150 words)	As large language models (LLMs) grow in size, running them efficiently has become a major engineering challenge. This project aims to build a scalable and cost-effective LLM inference system using the vLLM framework. We will start by deploying open-source models like Gemma on our local lab server to understand the baseline architecture, memory management, and precision tuning. Once validated, we will containerize the system using Docker and migrate it to a cloud environment. There, we will implement distributed inference across multiple GPUs to optimize throughput and latency. By benchmarking performance and resource costs, this project aligns with the DDPS Lab's focus on cloud optimization. Ultimately, we want to design a practical MLOps pipeline that serves LLMs reliably without excessive infrastructure costs.
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1. Project background and goals

[Background]

Running large language models (LLMs) requires massive computational power. A simple Python script is not enough to handle concurrent API requests in a real-world service, as it often leads to memory bottlenecks and poor resource utilization. To solve this, frameworks like vLLM, which use PagedAttention to manage memory efficiently and maximize throughput, have quickly become the industry standard for serving models.

[Goals]

The main goal of this project is to build a distributed inference system that can serve open-source LLMs stably and cost-effectively. We will take a two-step approach. First, we will use the DDPS Lab's local P100 GPU server to test the vLLM architecture, set up an API server, and tune parameters like data types to fit the hardware. Second, we will expand this setup to a public cloud environment using multi-GPU instances. By applying tensor and pipeline parallelism, we aim to find the optimal balance between inference speed and operational cost.

2. Project content

To ensure the system scales properly, we will implement and optimize the project in the following phases:

1. Local Environment Setup & Baseline Testing:We will start by allocating a single GPU on the DDPS lab server. We will load lightweight open-source models (like Gemma-2b), adjust the precision (dtype=half) to accommodate the P100's compute

capability, and run an OpenAI-compatible API server to test basic text generation and API communication.

2. Containerization & Security:To ensure the system can be deployed anywhere without version conflicts, we will package the vLLM environment using PyTorch Docker images. During this phase, we will resolve dependency issues (such as transformer library conflicts) and implement API key authentication to secure the endpoints.

3. Cloud Migration & Distributed Inference:We will migrate the Dockerized system to a cloud platform (e.g., AWS or GCP). Here, we will provision multi-GPU instances and set up distributed inference, splitting model weights across multiple GPUs to handle larger models and heavier user traffic.

4. Benchmarking & Optimization:We will conduct load testing to measure system latency and throughput (tokens/sec). We will compare the single-GPU local setup with the multi-GPU cloud setup to analyze cost-efficiency and identify the best resource allocation strategies.

3. Plan

5. March - April : Literature review on vLLM architecture. Set up SSH access to the DDPS lab server. Test local API serving and precision tuning on a single GPU.
6. May - June : Containerize the vLLM setup using Docker. Implement secure API communication and backend integration.
7. July - August : Provision cloud resources and migrate Docker containers. Set up the initial distributed inference architecture on a multi-GPU cloud instance.
8. September - October : Conduct traffic load testing. Benchmark performance and cost differences between the local and cloud environments.
9. November : Analyze benchmarking data and resolve system bottlenecks. Draft the final graduation thesis and prepare a manuscript for conference submission.
10. December : Finalize the report and present the capstone project.

(Role Division: Min Jihong - System architecture design, vLLM local and cloud deployment / Jeong Byeongyun - Docker containerization, API security setup, and performance benchmarking.)

4. Conclusion

[Technical Impact & Scalability]

This project goes beyond simply running a model; it provides a practical reference for scaling an AI system from a constrained local server to a high-performance cloud environment. By combining Docker and vLLM, the resulting architecture can be easily adapted by companies looking to deploy open-source LLMs cost-effectively in a cloud-native environment.

[Usefulness & Career Connection]

Through this project, we will gain hands-on experience with MLOps, system engineering, and cloud resource optimization. Solving real-world dependency conflicts and memory constraints while exposing a model as a production-ready API will significantly improve our problem-solving skills. These experiences will build a strong technical foundation for our future careers, preparing us to work as data scientists or ML engineers in professional sports front offices or global tech companies, where serving large-scale data efficiently is a top priority.

5. References

1. Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., ... & Stoica, I. (2023). "Efficient Memory Management for Large Language Model Serving with PagedAttention." *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP '23)*.

2. Blancas, E. (2024). "Deploying vLLM: a Step-by-Step Guide." *Ploomber Blog*.

3. Lee, S., Heo, Y., Park, S., and Lee, K. (2022). "PROFET: PROFiling-based CNN Training Latency ProphET for GPU Cloud Instances." *IEEE International Conference on Big Data*.

4. Choi, J., Park, S., and Lee, K. (2022). "All-You-Can-Inference: Serverless DNN Model Inference Suite." *Eighth International Workshop on Serverless Computing held with ACM/IFIP Middleware*.